COMPUTER ENGINEERING

QUEZON CITY

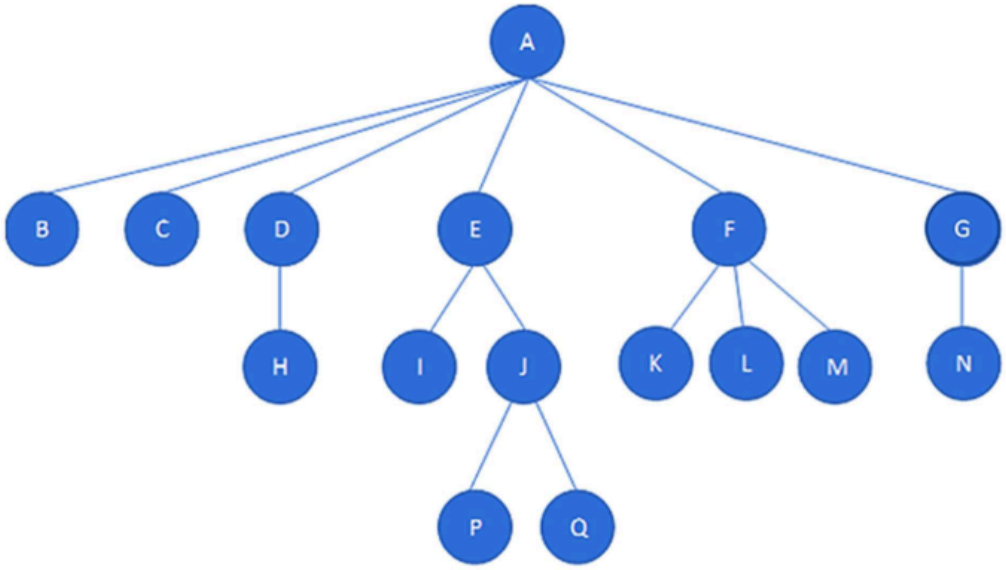
COURSE CODE/TITLE:	CPE 010-Data Structures and Algorithms	**SECTION:**	CPE21S4 -
DATE PERFORMED:	9/18/26	**DATE SUBMITTED:**	9/18/26
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ACTIVITY NO. 9.1 [Tree ADT]

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Create C++ code to implement a general tree, binary tree and binary search tree.



Task 1: Create code in C++ that will create a tree as shown in the figure above. Use linked lists as the internal representation of this tree. **Indicate your code screenshot and comments in table 9-1.**

Task 2: Complete the following table:

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

```
#ifndef BST_H
#define BST_H
```

```cpp
template <typename T>
class BinarySearchTree{
    private:
        T key;
        // pointers:
        BinarySearchTree<T>* left;
        BinarySearchTree<T>* right;

    public:
        //constructor:
        BinarySearchTree<T>(T value);


};

template <typename t>
BinarySearchTree<T>::BinarySearchTree(T value){
    key = value;
    left = nullptr;
    right = nullptr
}

template <typename T>
void BinarySearchTree<T>::insert(T value){
    if(value < key){
        //do something
        if(left == nullptr){
            left = new BinarySearchTree<T>(value);
        }else{
            left->insert(value);
        }
    }
    else if (value > key ){
        if(right == nullptr){
            right = new BinarySearchTree<T>(value);
        }else{
            right->insert(value);
        }
```

```
    }
}

template <typename T>
void BinarySearchTree<T>::preOrder(){
    std::cout<<key"";
    if(left !=nullptr){
        left->preOrder();
}
```

Main

```
#include <iostream>
#include "BST.h"


int main(){
    //create the root node:
    BinarySearchTree<int> tree(20);

    tree.insert(10);
    tree.insert(5);
    tree.insert(70);
    tree.insert(80);
    tree.insert(40);
    tree.insert(50);

    std::cout<<"preOrder: ";
    tree.preOrder();
    std::cout<<std:endl;

    std::cout<<"inOrder: ";
    tree.inOrder();
    std::cout<<std:endl;

    std::cout<<"postOrder: ";
```

```
    tree.inOrder();
    std::cout<<std:endl;



    return 0;
}
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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